Module 3.3 — Change Summary for Review

File: `3.3 Tuning Tree-Based Models.ipynb` (the plain, no-suffix lecture — the file we agreed to keep as canonical)

Status: Change applied, awaiting review. `3.3 Tuning Tree-Based Models Using f1.ipynb` has **not** been deleted yet — that happens only after this is approved.

What changed

Only one cell was touched — the data-setup cell near the top of the notebook. Everything else (Decision Tree / Random Forest / XGBoost tuning, the early-stopping section, the final model-save cell) is untouched.

Before

```
import numpy as np
import pandas as pd
import warnings
warnings.filterwarnings('ignore')

import plotly.graph_objects as go
from plotly.subplots import make_subplots

from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

from sklearn.model_selection import (GridSearchCV, RandomizedSearchCV,
                                     StratifiedKFold, cross_val_score)
from sklearn.metrics import roc_auc_score, make_scorer, f1_score
import time

RANDOM_STATE = 42
np.random.seed(RANDOM_STATE)

# Load and prepare data (same as 3.2)
train_df = pd.read_csv(f'{project_path}{data_filepath}training.csv')
test_df = pd.read_csv(f'{project_path}{data_filepath}testing.csv')
train_df['DEPARTED'] = (train_df['SEM_3_STATUS'] != 'E').astype(int)
test_df['DEPARTED'] = (test_df['SEM_3_STATUS'] != 'E').astype(int)

numeric_features =
['HS_GPA','HS_MATH_GPA','HS_ENGL_GPA','UNITS_ATTEMPTED_1','UNITS_ATTEMPTED_2

'UNITS_COMPLETED_1','UNITS_COMPLETED_2','DFW_UNITS_1','DFW_UNITS_2','GPA_1',
    'DFW_RATE_1','DFW_RATE_2','GRADE_POINTS_1','GRADE_POINTS_2']
categorical_features =
['RACE_ETHNICITY','GENDER','FIRST_GEN_STATUS','COLLEGE']

train_enc = pd.get_dummies(train_df[numeric_features +
categorical_features],
                           columns=categorical_features, drop_first=True)
test_enc = pd.get_dummies(test_df[numeric_features +
categorical_features],
                          columns=categorical_features, drop_first=True)
train_enc, test_enc = train_enc.align(test_enc, join='left', axis=1,
fill_value=0)
train_enc = train_enc.fillna(train_enc.median())
```

```
test_enc = test_enc.fillna(train_enc.median())

X_train, y_train = train_enc, train_df['DEPARTED']
X_test, y_test = test_enc, test_df['DEPARTED']

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE)
print(f"Data loaded: {X_train.shape[0]:,} training, {X_test.shape[0]:,}
testing samples")
```

After

```
import numpy as np
import pandas as pd
import joblib, os
import warnings
warnings.filterwarnings('ignore')

import plotly.graph_objects as go
from plotly.subplots import make_subplots

from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

from sklearn.model_selection import (GridSearchCV, RandomizedSearchCV,
                                     StratifiedKFold, cross_val_score)
from sklearn.metrics import make_scorer, f1_score
import time

RANDOM_STATE = 42
np.random.seed(RANDOM_STATE)

ARTIFACT_DIR = f'{project_path}{course3_models}'   # artifacts persist
here
os.makedirs(ARTIFACT_DIR, exist_ok=True)

# Load and prepare data (same as 3.2)
train_df = pd.read_csv(f'{project_path}{data_filepath}training.csv')
test_df = pd.read_csv(f'{project_path}{data_filepath}testing.csv')
train_df['DEPARTED'] = (train_df['SEM_3_STATUS'] != 'E').astype(int)
test_df['DEPARTED'] = (test_df['SEM_3_STATUS'] != 'E').astype(int)

numeric_features =
['HS_GPA','HS_MATH_GPA','HS_ENGL_GPA','UNITS_ATTEMPTED_1','UNITS_ATTEMPTED_2

'UNITS_COMPLETED_1','UNITS_COMPLETED_2','DFW_UNITS_1','DFW_UNITS_2','GPA_1',
    'DFW_RATE_1','DFW_RATE_2','GRADE_POINTS_1','GRADE_POINTS_2']
categorical_features =
['RACE_ETHNICITY','GENDER','FIRST_GEN_STATUS','COLLEGE']

train_enc = pd.get_dummies(train_df[numeric_features +
categorical_features],
                           columns=categorical_features, drop_first=True)
test_enc = pd.get_dummies(test_df[numeric_features +
```

```
categorical_features],
                            columns=categorical_features, drop_first=True)
train_enc, test_enc = train_enc.align(test_enc, join='left', axis=1,
fill_value=0)

# --- Capture medians BEFORE filling so they persist; apply TRAIN medians
to both ---
train_medians = train_enc.median()
train_enc = train_enc.fillna(train_medians)
test_enc  = test_enc.fillna(train_medians)

X_train, y_train = train_enc, train_df['DEPARTED']
X_test, y_test = test_enc, test_df['DEPARTED']

feature_columns = list(train_enc.columns)

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE)
print(f"Data loaded: {X_train.shape[0]:,} training, {X_test.shape[0]:,}
testing samples")

# --- Persist artifacts for 3.4 ---
joblib.dump(feature_columns, f'{ARTIFACT_DIR}feature_columns.pkl')
joblib.dump(train_medians,   f'{ARTIFACT_DIR}train_medians.pkl')
print(f"Saved {len(feature_columns)} feature columns +
{len(train_medians)} medians → {ARTIFACT_DIR}")
```

What the two changes do

1. **Added the missing helper-file save** (`feature_columns.pkl`, `train_medians.pkl`). This is the piece `3.4 Evaluating Tree-Based Models.ipynb` needs that only the "Using f1" file used to produce. This notebook now saves it too, alongside the three tuned `.joblib` models it already saved — so a single run of this one notebook produces everything 3.4 needs.
2. **Removed the unused `roc_auc_score` import.** It was imported but never called anywhere in this file, so the file is now unambiguously F1-only.

Why the plain file, not "Using f1"

Checked every AUC/F1 usage in both files line by line. Both tune on `scoring='f1'` — that part was already fine in both. The real difference:

Model	”Using f1” reports	plain file reports
Decision Tree	Best CV F1 + **Test AUC**	Best CV F1 + **Test F1**
Random Forest	Best CV F1 + **Test AUC**	Best CV F1 + **Test F1**
XGBoost	Best CV F1 + **Test AUC**	Best CV F1 + **Test F1**
XGBoost (early stopping)	**Test AUC**	**Test F1**
Final summary table	Has a `ROC-AUC` column	F1 / Precision / Recall only

Despite its name, “Using f1” tunes on F1 but reports the held-out **test-set** result in AUC every single time — six separate, real, executed `roc_auc_score` calls. The plain file is F1-consistent end to end: tunes on F1, reports test performance in F1, with zero AUC anywhere (until this edit removed even the one unused import).

Verification status

No data files available locally yet, so this has been verified **statically only**:

- Every variable (`feature_columns`, `train_medians`, `joblib`, `os`, `ARTIFACT_DIR`) is defined before it’s used.
- Zero remaining `roc_auc_score` references anywhere in the file.
- The saved filenames exactly match what `3.4 Evaluating Tree-Based Models.ipynb` already loads.

Not yet confirmed by actually running the notebook end-to-end against real data.

Next step

Once this is approved, `3.3 Tuning Tree-Based Models Using f1.ipynb` will be removed so only one canonical 3.3 file remains.